

w & $b \rightarrow$ Initialize

first pass

compute loss

loss

$\rightarrow$ loss

should
minimize

$$\frac{\partial L}{\partial w}, \frac{\partial L}{\partial b}$$

Optimization $\rightarrow$ update w & b

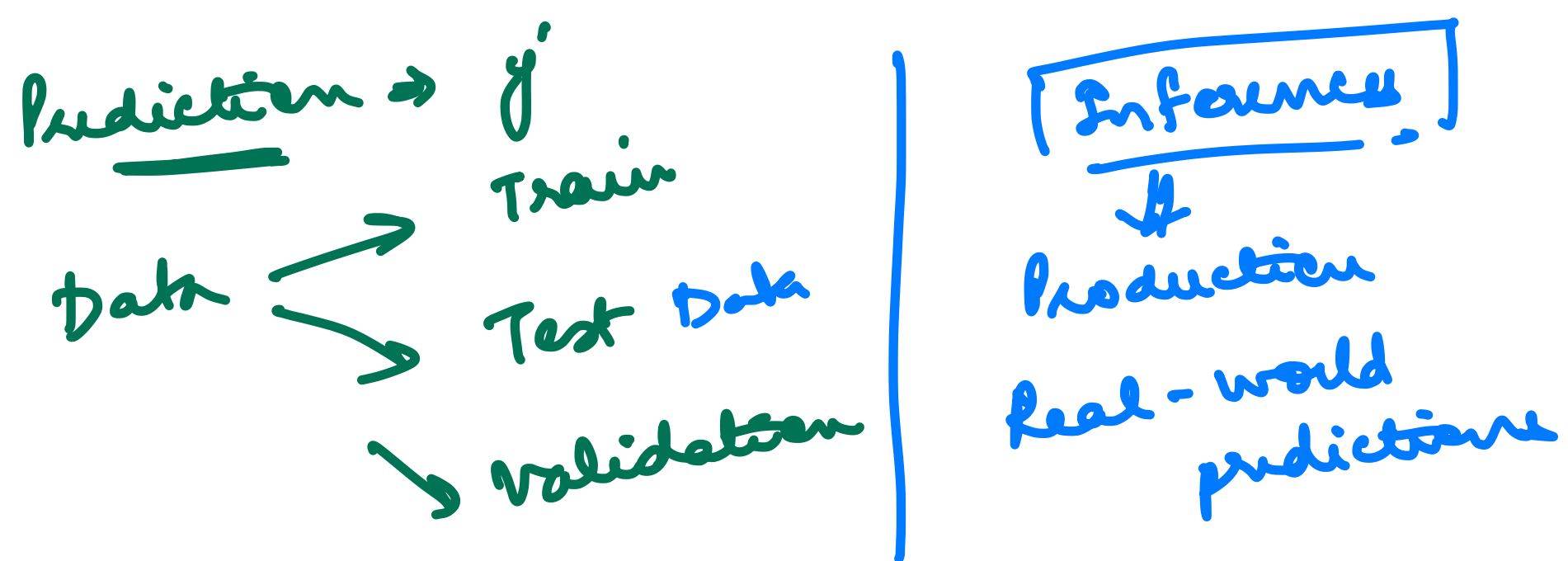
loss
convergence

Stability

unstable weight updates

vanishing gradient problem

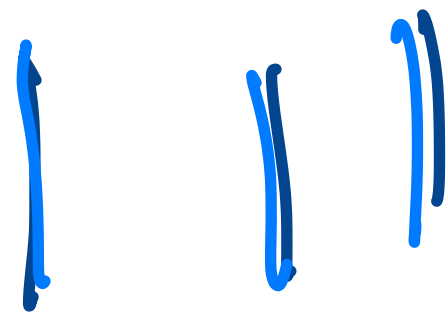
Exploding gradient problem



Layers

- $\rightarrow$ Dense $\rightarrow$ fully connected layers
- $\rightarrow$ Flatten $\rightarrow$ 0 params
 $\rightarrow$ 1D
- $\rightarrow$ Conv2D $\rightarrow$ Detecting patterns/features
Spatial
- $\rightarrow$ Conv Transpose $\rightarrow$ Upsampling
- $\rightarrow$ Reshape $\rightarrow$ Change dimensions
0 params

Sequential model → starting point



Dense.

→ Embedding layer → content/relationships

Overfitting ?? → Training Accuracy is good
Testing Accuracy is X

→ Memorize for Training Data
→ Randomness is needed.

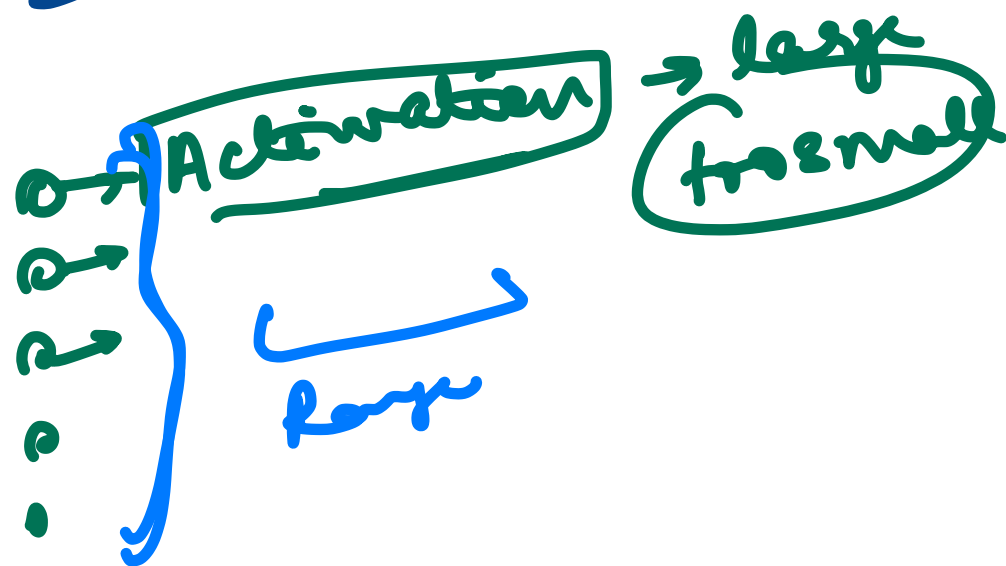


Drop Neurons

Dropout layers → Improving generalisation
→ ignored during inference

Testing is also type of inference
Not all inference is testing

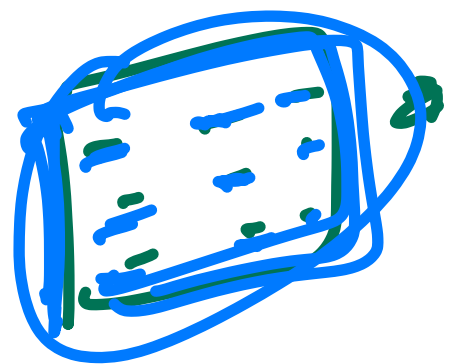
Batch Normalization layer → mean, std



faster convergence?

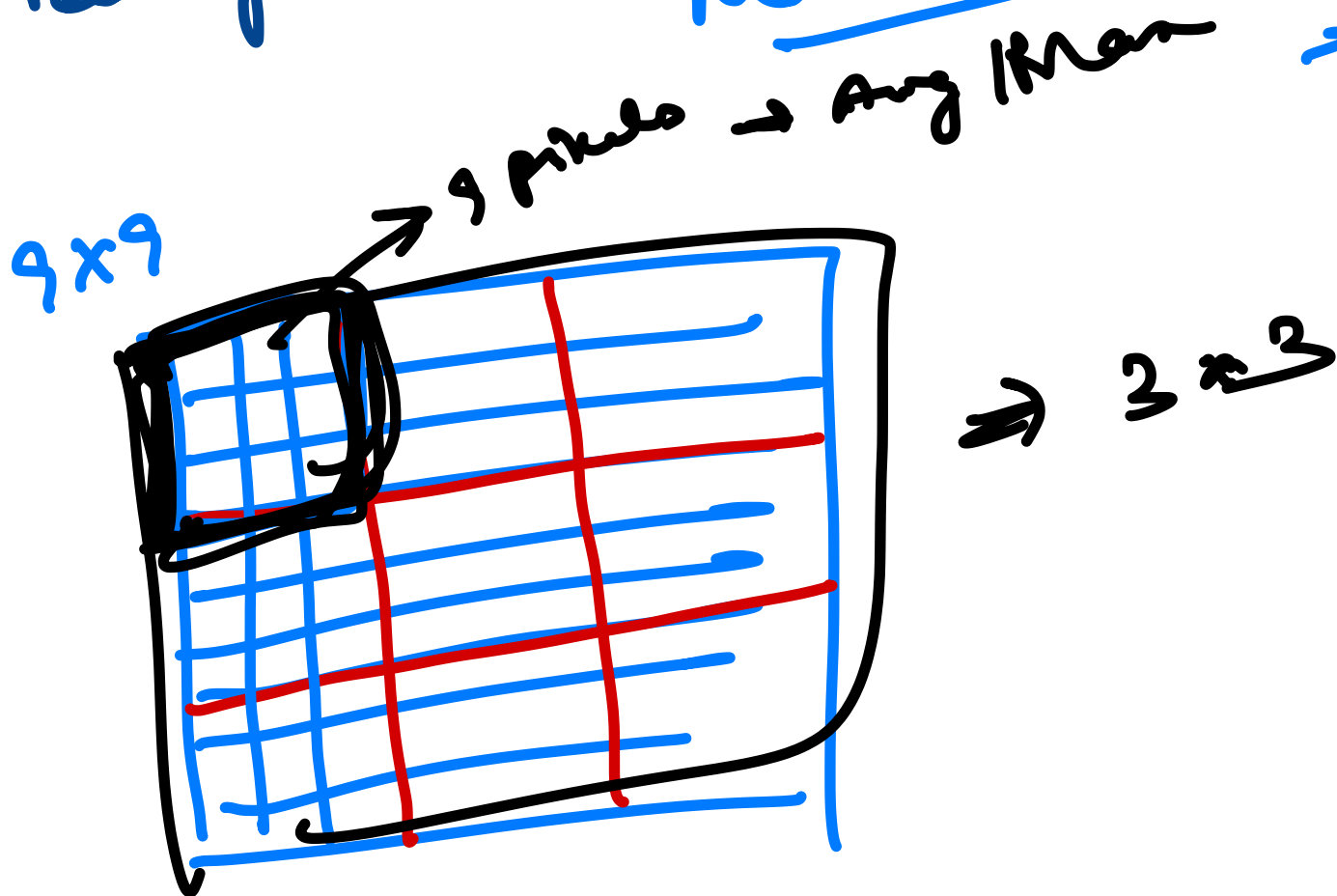
Recurrent layers $\Rightarrow$ RNN
 $\Rightarrow$ LSTM, GRU

Attention layers $\Rightarrow$



Blur. No detection
Downsampling of image
Resolution $\downarrow$

Pooling layers $\Rightarrow$ Avg Pooling
Max Pooling $\Rightarrow$ Downsampling
 $\rightarrow$ Not Memory



ANN

neurons

connections $\rightarrow$ wts

bias

Activation
Func

CNN

filter / kernels

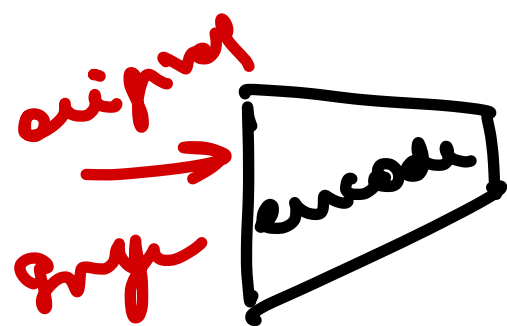
elements in
filter

bias

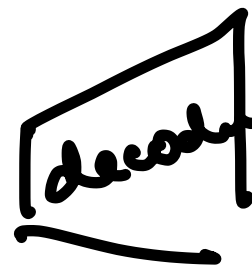
Activation
Func

Feature Map

o/p $\rightarrow$ Activation



$[]$
vector
latent space /
bottleneck

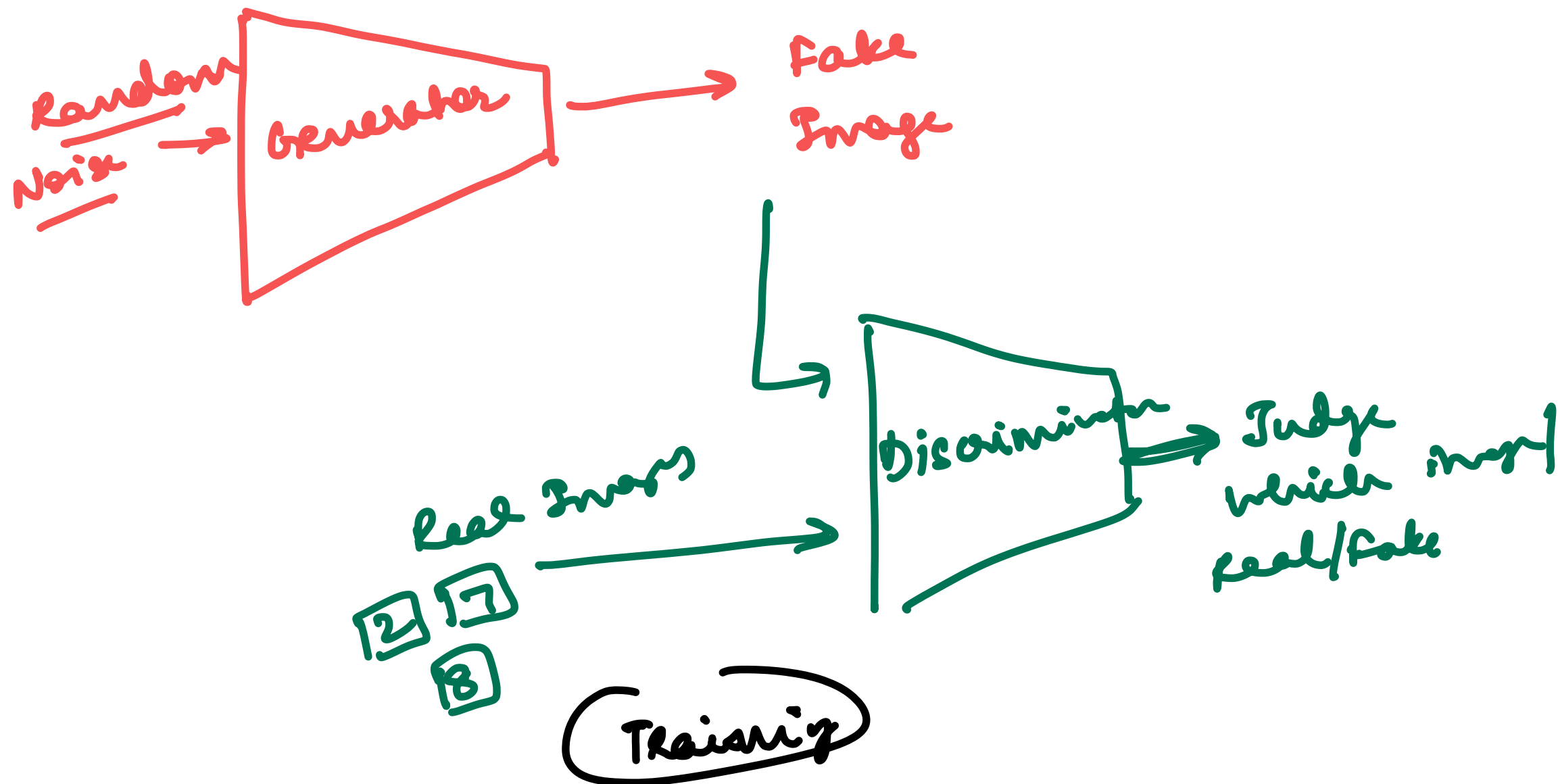


$\rightarrow$ Reconstructed
Image

$$\text{Loss} = \left\{ \frac{\text{Original Image}}{\text{Reconstructed Image}} \right\}$$

Goal → Generate new GAN images

2 Models → enemies
→ Fight



① Generator generated
fake image from noise

② Discriminator evaluate
→ Real Image → 1 (ideal o/p)
→ Fake Image → 0 (ideal o/p)

Goal of generator
 $\text{Disc}(\text{Fake Image}) = 1$

Goal of Disc

Real Image → 1

Fake Image → 0

winner, loser

$\text{Disc}(\text{Fake Image}) \Rightarrow 0$

Disc ✓

Gen X

W 2 b

Gen ✓

$\text{Disc}(\text{Fake Image}) \Rightarrow 1$

Disc

X

W 2 b

(Testing) / Inference

Generator

No role of discriminator

→ Equilibrium →
Generator loss in minimum
Discriminator loss is 0.5

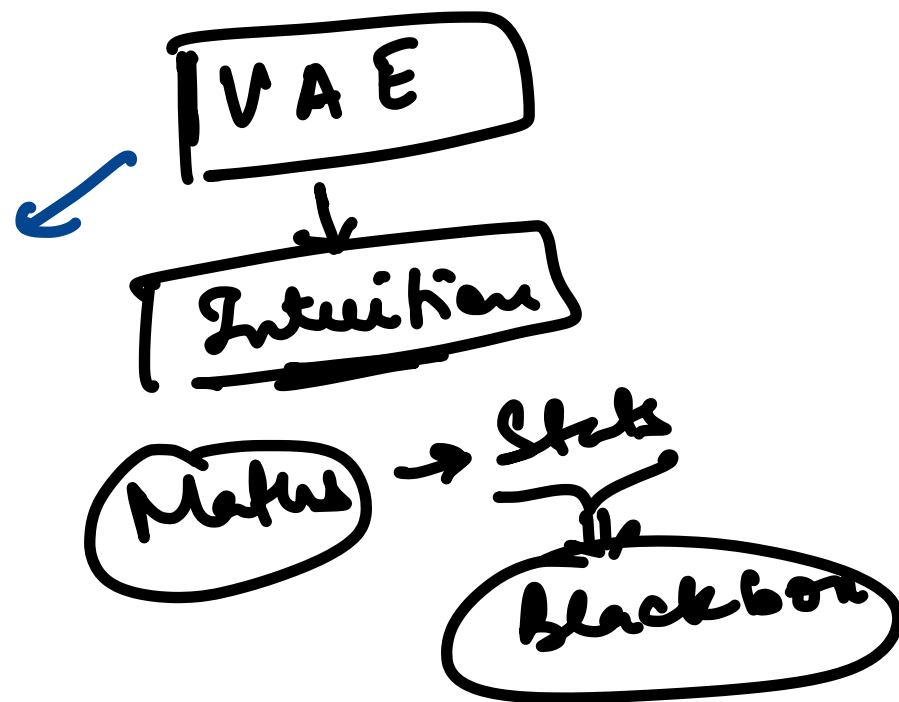
Low Disc loss → Disc strong (very)
→ generator → weak
→ Disc distinguish

High Disc loss → Disc struggling
→ generator → v. strong } mode collapse.

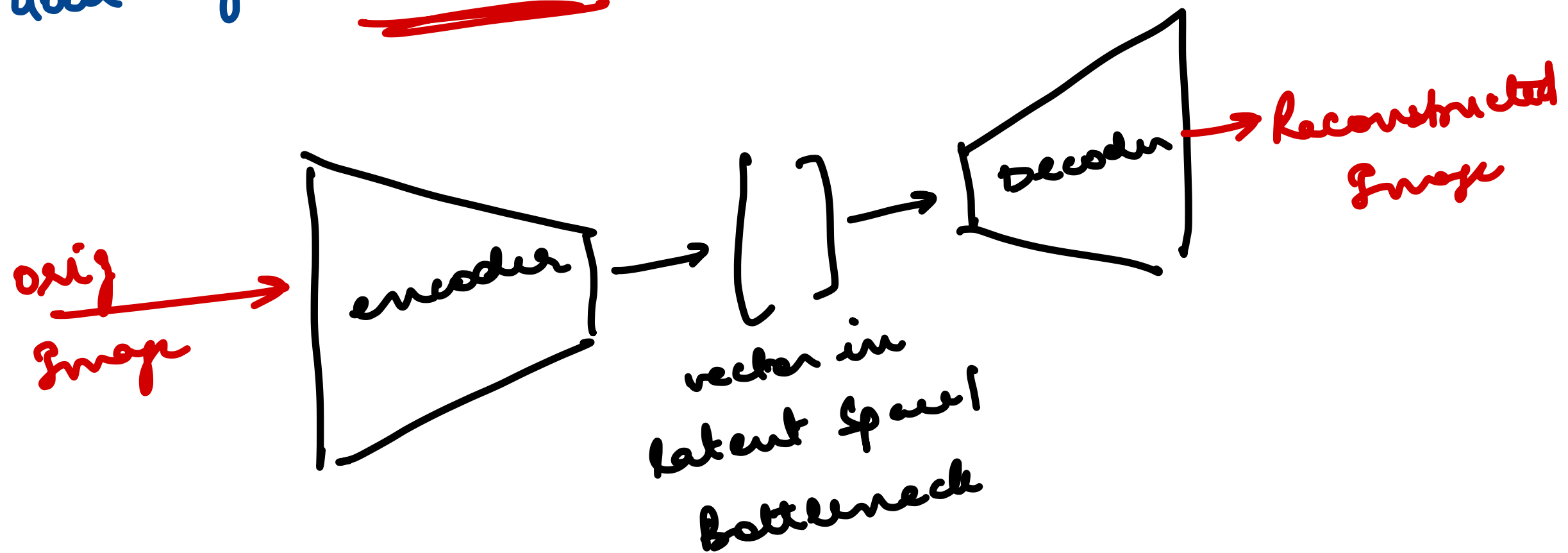
Dis loss
stable no.5 $\Rightarrow$ Ideal Scenario

Gen loss ≈ 0
Disc loss $\approx 0.5 \rightarrow$ confused
 $\rightarrow$ not able to distinguish

variational
autoencoder



Goal $\rightarrow$ generate New Image



loss $\rightarrow$ orig Image, Reconstructed Image

human face $\rightarrow$ Cat Face / Dog Face
Cat $\rightarrow$ Bear / Elephant

Similar New Image
Not be a random Image

Human $\rightarrow$ $\left\{ \begin{array}{l} \text{Red hair} \\ \text{Specs / No Specs} \\ \text{Eye} \end{array} \right\}$ Human face

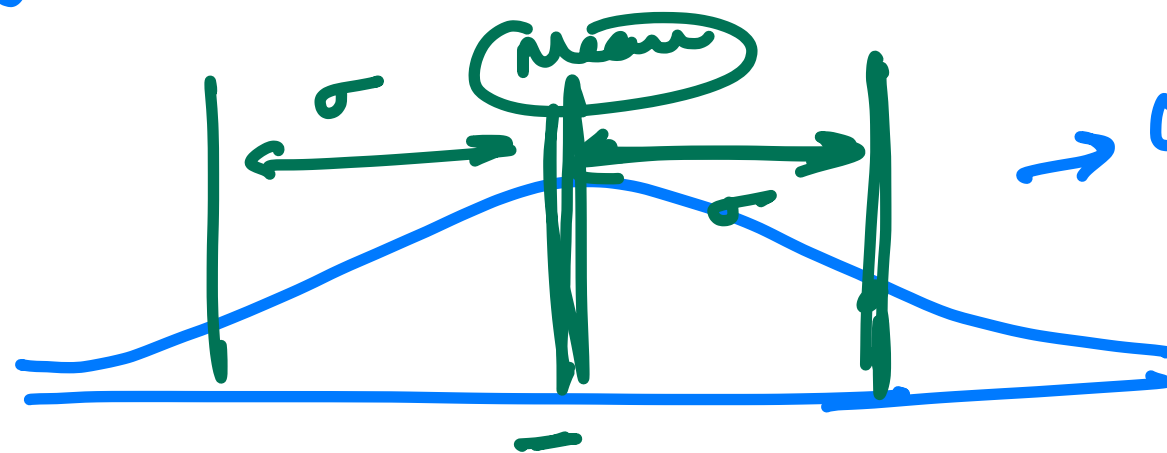
New Image $\rightarrow$ Balance

Random Cat Elephant

Same $\rightarrow$ Slightly different / new

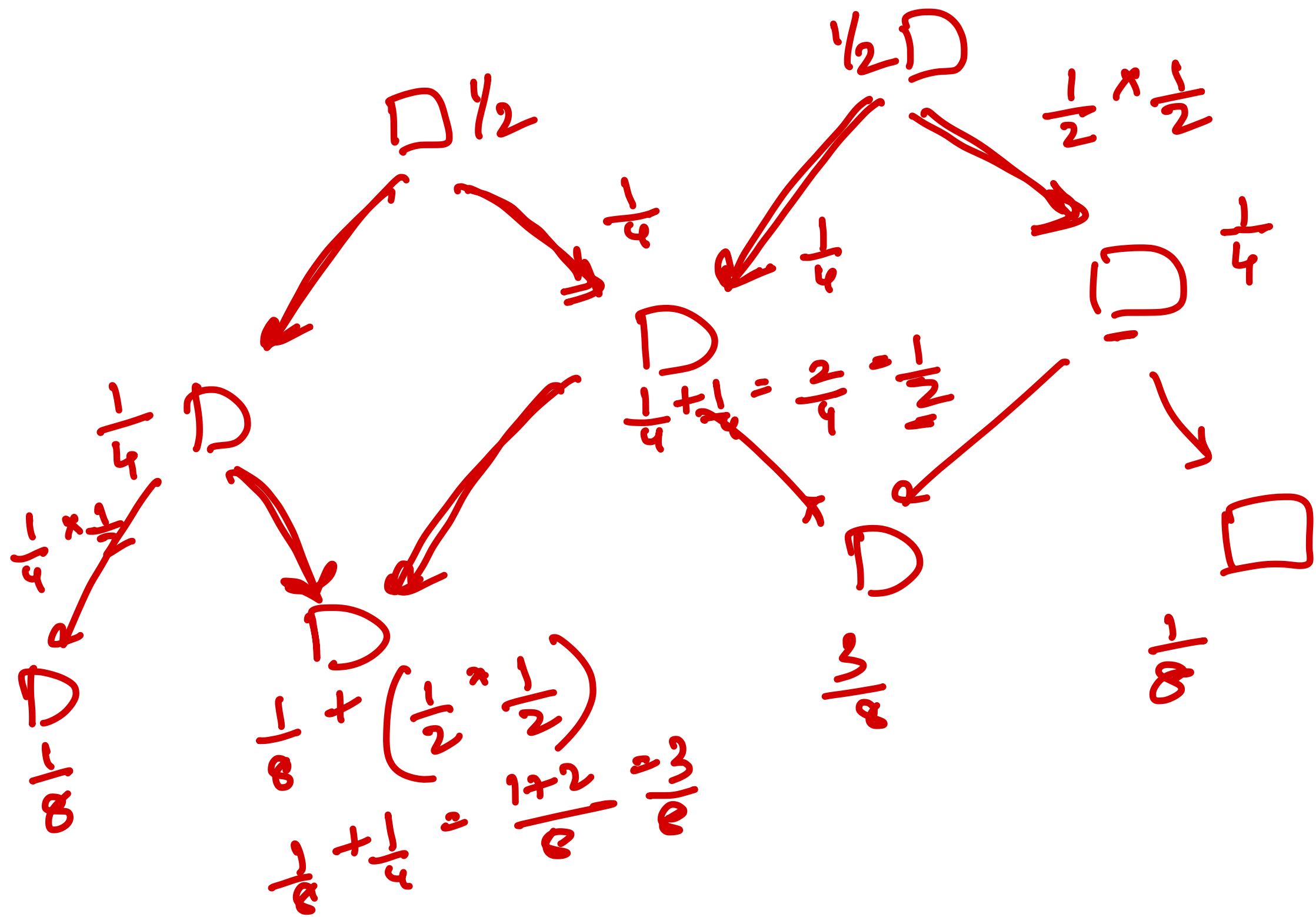
Morphing face

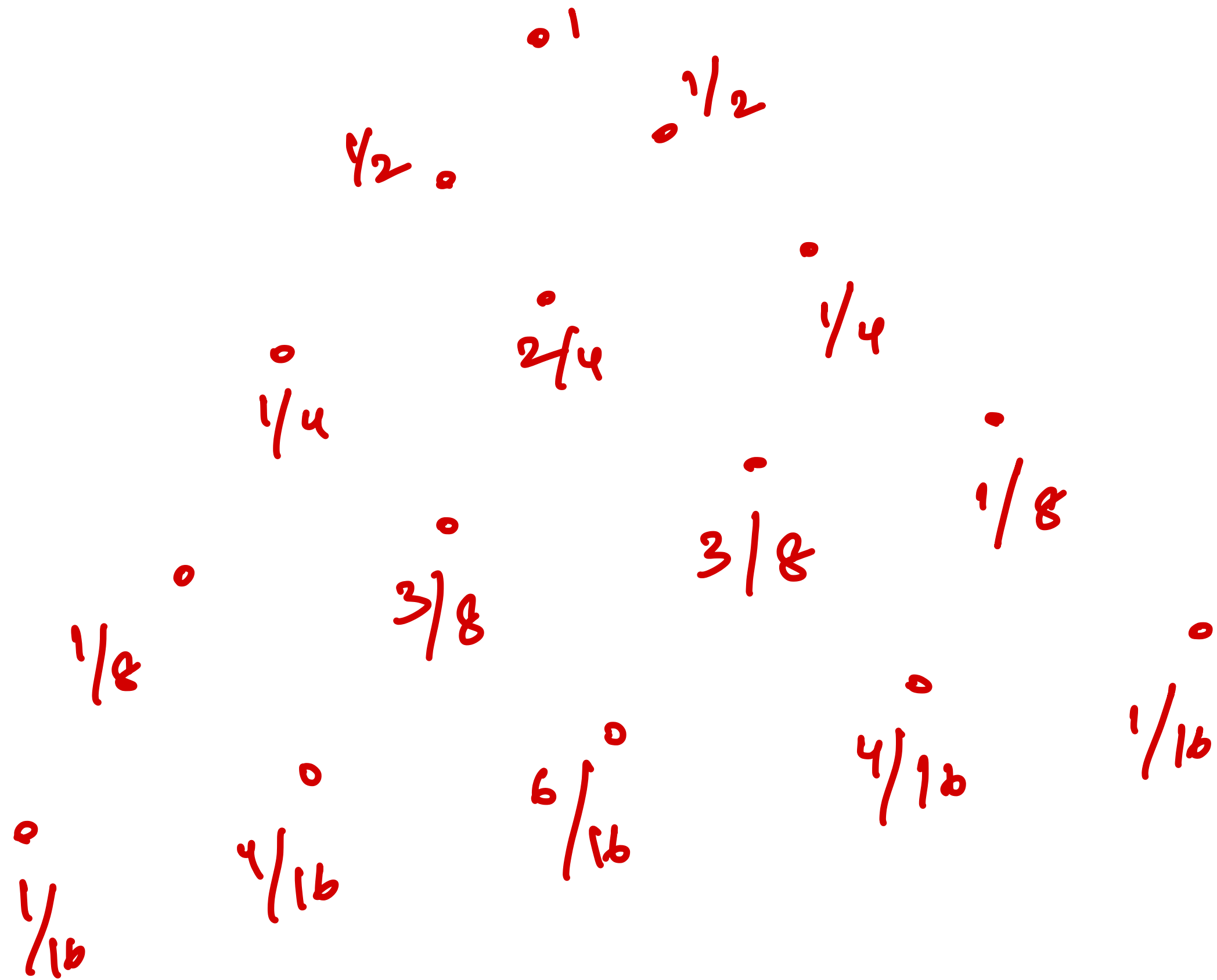
Alia Bhatt $\rightarrow$ Random face



$\rightarrow$ Gaussian Distribution / Normal Distribution

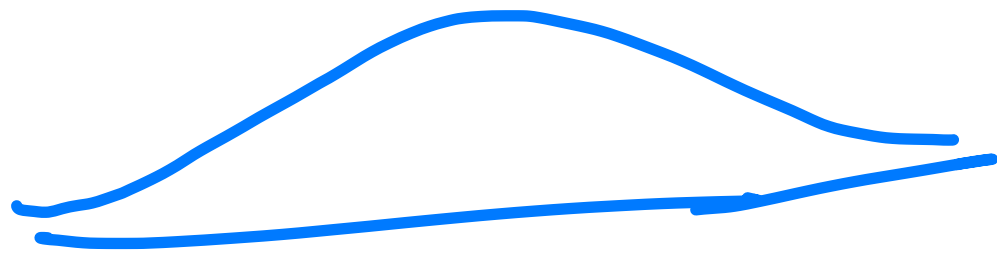
10



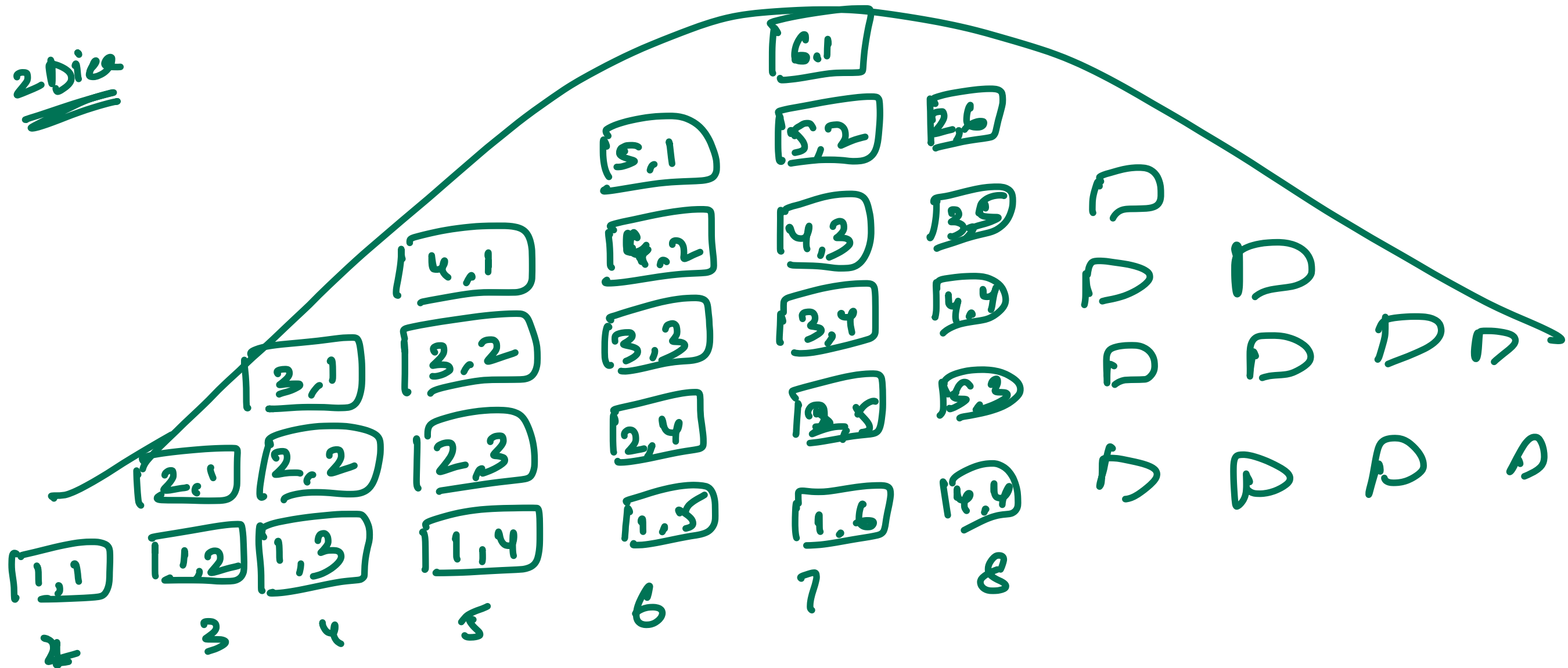


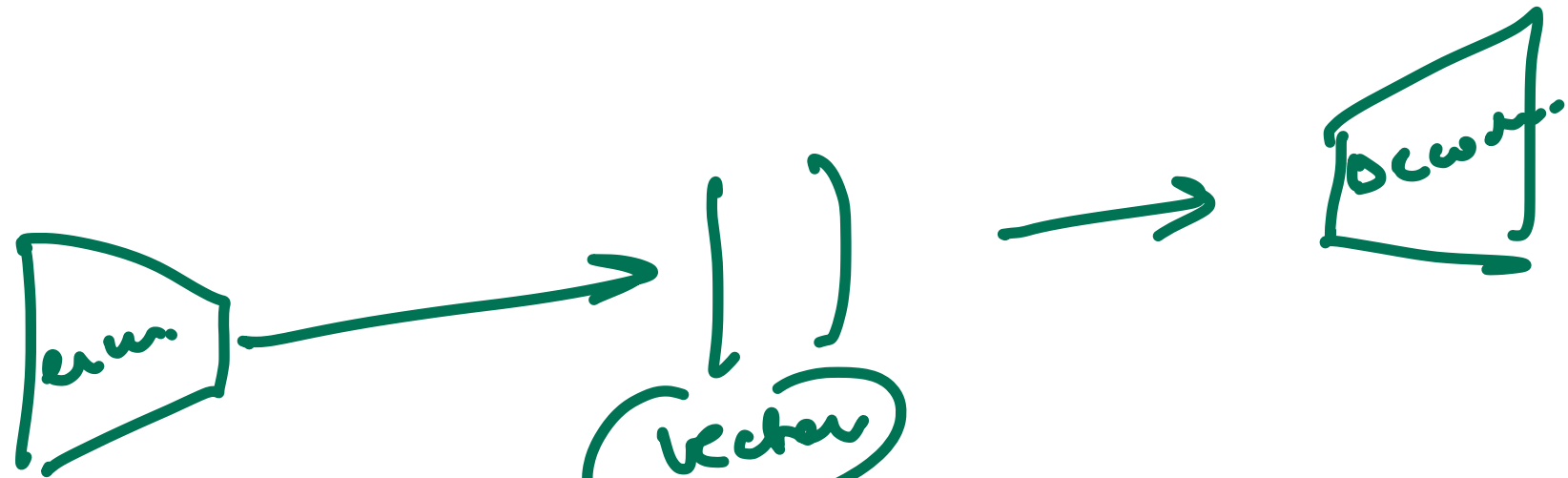
Central limit Theorem

1000 people $\rightarrow$ heights

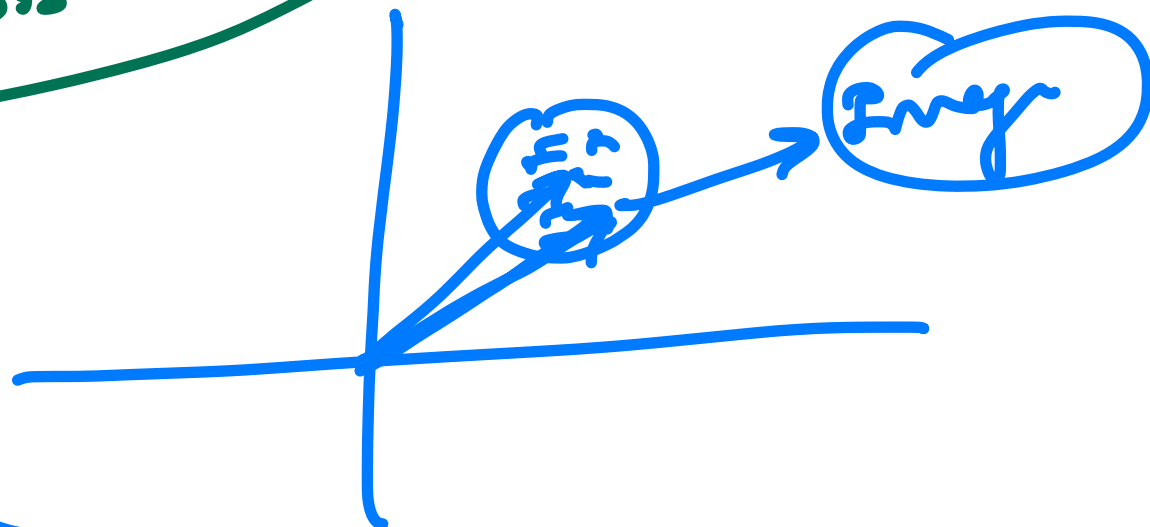


2 Dice





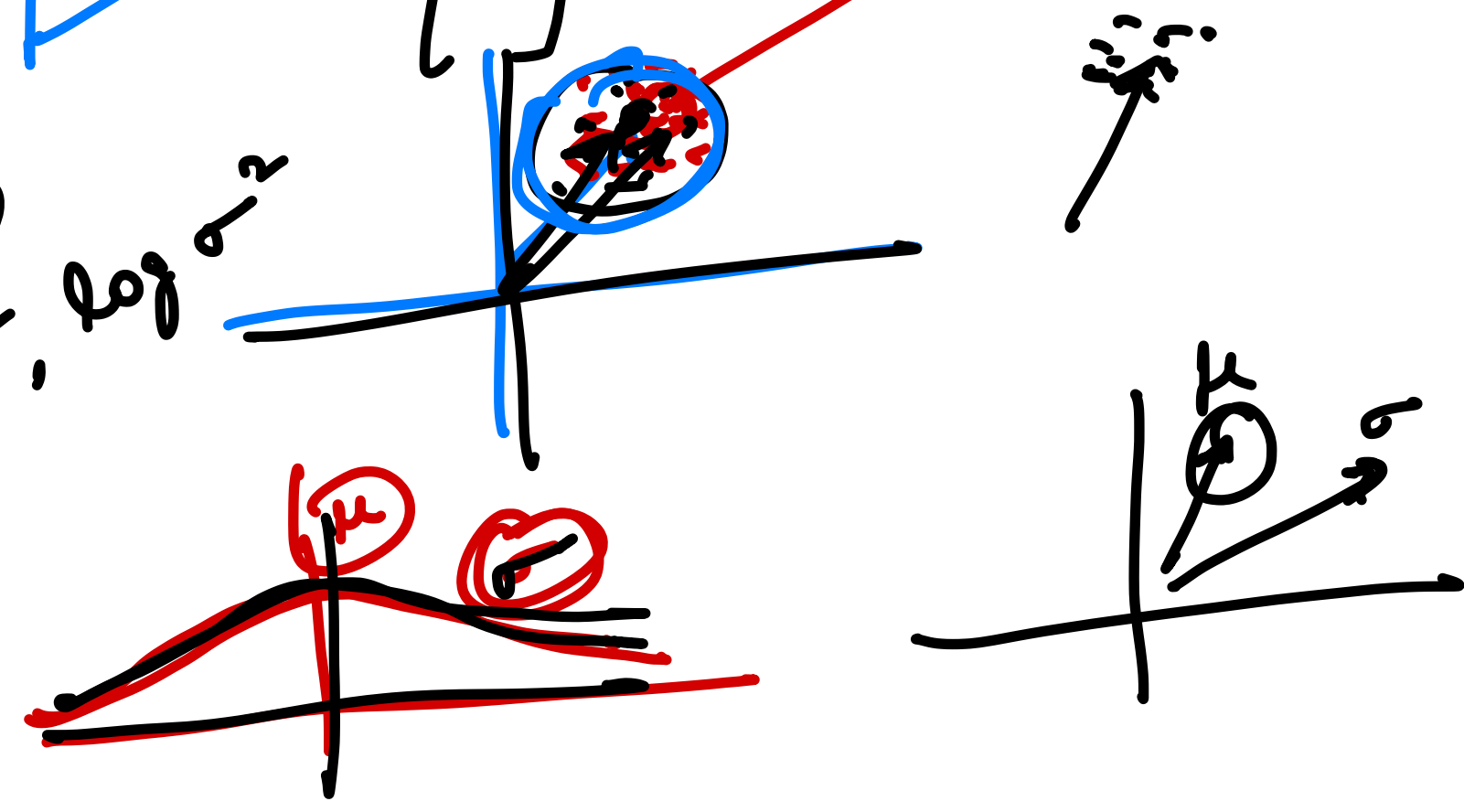
Similar
Probab Distributa



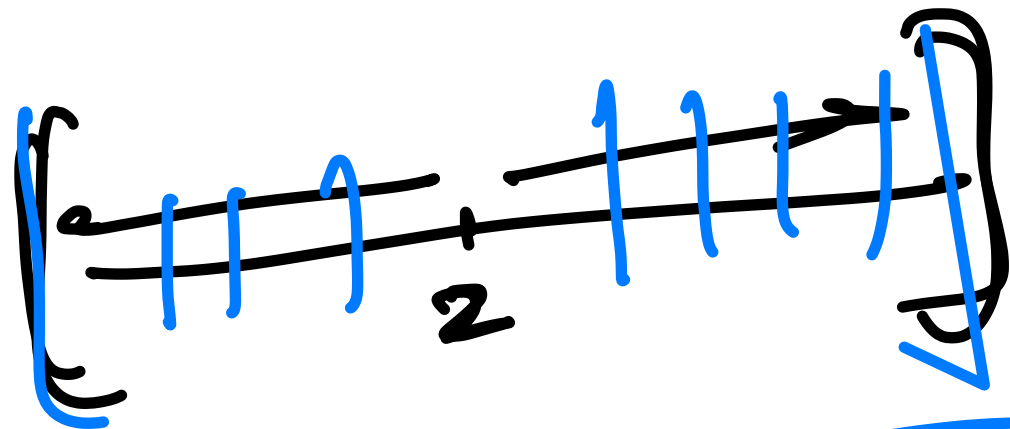
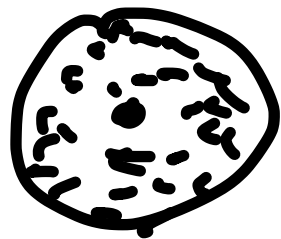
2 vectors $\rightarrow$ Mean $\rightarrow \frac{v_1 + v_2}{2}$



μ , σ^2 , $\log \sigma^2$



Distribution $\rightarrow$ encode $\rightarrow (\mu, \sigma)$ $\rightarrow$ [] $\rightarrow$

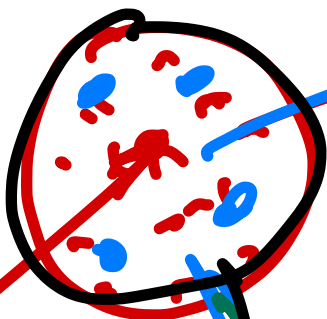


2 1 3

2+1
2-1

2+2 2+3
2-2 2-3

μ, σ
param



Death

New
Inv

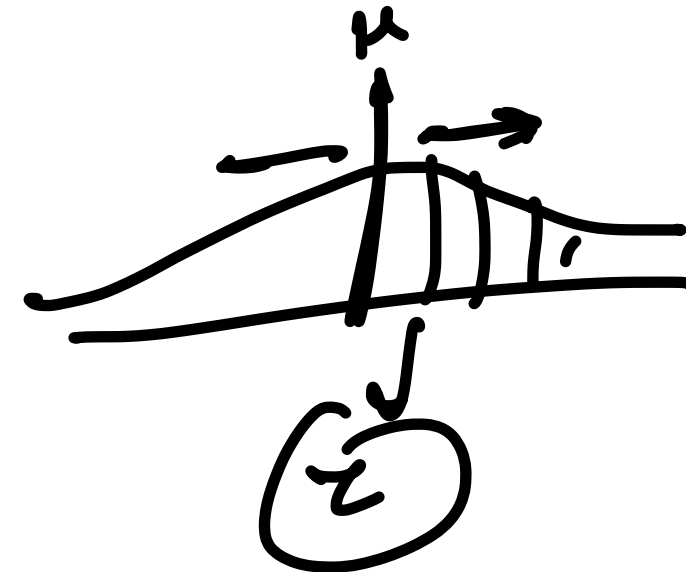
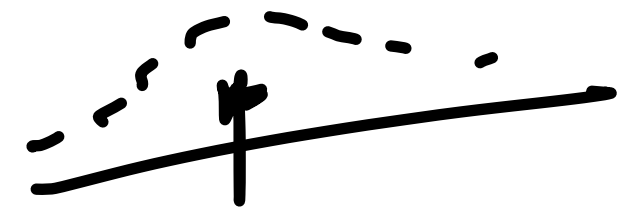
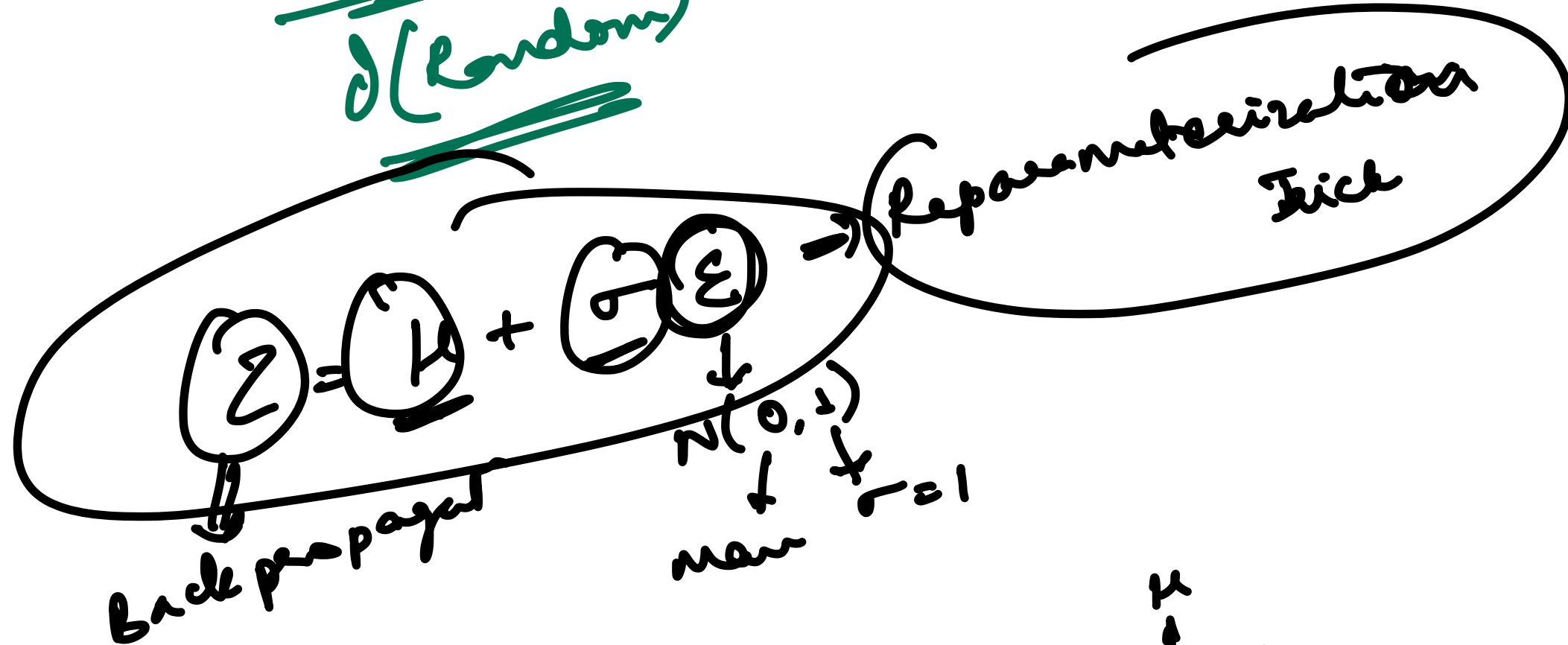
Sampling

Sample from
distribution

Loss

w, b
optimization

$\delta(\text{loss})$
 $\delta(\text{random})$



VAE

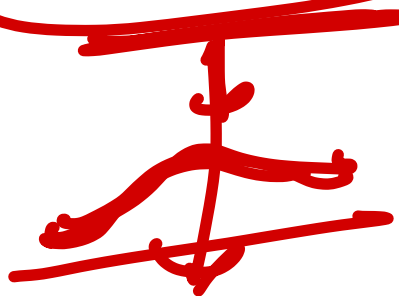
Loss

Reconstruction loss

$\Downarrow$

Images is similar

+ KL Divergence



How far our
distribution is from
other distribution

